Real-Time Sign Language Detection and Translation System



Abstract

Sign language is the primary mode of communication for millions of individuals who are Deaf or hard of hearing, yet the vast majority of the hearing population has no working knowledge of it. This gap creates persistent barriers in everyday interactions such as education, healthcare, banking, and public services. This project proposes a real-time sign language detection and translation system that uses a standard camera feed to recognize hand gestures and convert them into readable text and synthesized speech, without requiring any specialized wearable hardware such as sensor gloves. The system captures live video, extracts hand landmarks using a lightweight computer-vision pipeline, and classifies the resulting gesture sequence using a hybrid Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) model trained on a labeled sign vocabulary. Recognized signs are assembled into coherent sentences and passed to a text-to-speech engine so that a non-signing listener can understand the message audibly, while the recognized text is simultaneously displayed on screen for accessibility in noisy or quiet settings. The methodology emphasizes low-cost deployment on consumer-grade hardware, targeting near real-time latency and high classification accuracy on a curated set of common signs. The expected outcome is a working prototype that demonstrates two-way accessible communication between Deaf and hearing individuals, along with a quantitative evaluation of recognition accuracy, latency, and robustness under varying lighting and background conditions. The project aims to contribute a practical, extensible step toward affordable assistive communication technology.

I. Introduction

Communication is fundamental to social participation, education, and employment. For the global Deaf and hard-of-hearing community, sign language is not merely an alternative to spoken language but a complete linguistic system with its own grammar and structure. However, fluency in sign language remains rare among hearing individuals, and professional human interpreters are often unavailable, expensive, or difficult to schedule on short notice. Advances in computer vision and deep learning over the past decade have made it possible to interpret human gestures directly from video, opening the door to automated, low-cost communication aids. Early gesture-recognition systems relied on statistical models such as Hidden Markov Models applied to hand-tracked features, achieving reasonable results on small isolated-sign vocabularies but struggling to generalize to continuous, natural signing. More recent CNN-based approaches improved raw classification accuracy on static and dynamic gestures, and transformer- and sequence-based architectures have since pushed continuous sign language recognition and translation closer to practical deployment. Despite this progress, most high-accuracy systems remain confined to research settings, relying on specialized hardware, large annotated datasets, or computational resources that are not readily accessible for everyday, low-cost use. This project builds on that body of work by targeting a deployable, camera-only pipeline that runs on consumer-grade hardware while still delivering real-time performance.

A. Motivation

According to global health estimates, several hundred million people live with disabling hearing loss, and a significant subset rely on sign language as a first or preferred language. The absence of affordable, always-available translation tools forces Deaf individuals into avoidable dependency on interpreters or written notes for routine interactions — a delay and inconvenience that hearing individuals rarely experience. This creates measurable disadvantages in access to healthcare, education, and employment. A real-time, camera-based translation system running on an ordinary laptop or smartphone could reduce this dependency, lower the cost of accessible communication, and be deployed at points of frequent need such as reception desks, classrooms, and service counters. Solving this problem is therefore both a matter of technical interest and social impact.

B. Problem Statement

People who use sign language often face communication barriers when interacting with individuals who do not understand it. This project addresses that gap by designing a system that detects hand signs from a live camera feed and converts them into text and speech in real time, without requiring specialized wearable sensors or a human interpreter.

C. Objectives

The primary objective of this project is to design and implement a real-time, vision-based system that translates hand signs into text and speech using only a standard camera.

- Design and train a hand-gesture classification model covering an initial vocabulary of at least 30–50 common signs and the English alphabet.
- Achieve a gesture classification accuracy of at least 90% on a held-out validation set under normal indoor lighting.
- Achieve an end-to-end recognition latency of under 300 milliseconds per gesture so that translation feels conversational rather than delayed.
- Implement sentence-level assembly logic that groups sequential signs into readable phrases rather than isolated words.
- Integrate a text-to-speech module so recognized signs are output as both on-screen text and synthesized audio.

II. Literature Survey / Related Work

The following table summarizes representative work in automated sign language recognition, spanning classical statistical methods to modern deep-learning approaches.

S.No.	Author(s) & Year	Approach / Method Used	Limitation / Gap
1	Starner & Pentland (1995)	Used Hidden Markov Models with a colored-glove tracking setup to recognize American Sign Language sentences from video.	Required colored gloves and a constrained vocabulary; not robust to bare-hand, unconstrained backgrounds.
2	Pigou et al. (2015)	Applied deep convolutional neural networks to Italian sign gesture recognition from RGB-D (Kinect) video.	Depends on depth-sensing hardware (Kinect), limiting deployment on ordinary consumer cameras.
3	Koller et al. (2016)	Combined CNNs with HMMs for continuous, weakly-supervised sign language recognition on the RWTH-PHOENIX-Weather corpus.	High computational cost and reliance on large, domain-specific (weather broadcast) annotated corpora.
4	Camgoz et al. (2018)	Proposed neural sign language translation using an encoder-decoder architecture that maps sign video directly to spoken-language text.	Achieves strong translation quality but needs substantial training data and GPU resources, limiting lightweight deployment.
5	Rastgoo et al. (2021)	Survey of deep-learning-based sign language recognition methods, comparing CNN, RNN/LSTM, and hybrid	Identifies that most high-accuracy systems are evaluated offline and are not

		architectures across multiple datasets.	optimized for real-time, low-resource inference.
6	Li et al. — WLASL Dataset (2020)	Introduced the Word-Level American Sign Language (WLASL) video dataset and benchmarked I3D-based recognition models on it.	Large-scale word-level dataset improves benchmarking, but models trained on it are heavy for real-time, on-device use.

Across these works, a consistent gap emerges: systems that achieve high accuracy typically depend on specialized sensors (gloves, depth cameras) or heavy models unsuitable for real-time inference on affordable hardware, while lightweight systems tend to support only small, isolated-gesture vocabularies. This project targets that middle ground — a bare-hand, RGB-camera-only pipeline using efficient landmark-based features and a compact CNN-LSTM classifier, aiming for real-time performance without specialized hardware or large-scale infrastructure.

III. Proposed System / Methodology

A. System Overview

The proposed system replaces sensor-based or depth-camera-dependent pipelines with a bare-hand, RGB-only approach. A standard webcam captures live video, from which hand landmarks are extracted using a lightweight, pre-trained hand-tracking model. These landmarks — rather than raw pixels — form a compact feature representation that is robust to background clutter and lighting variation, and is efficient enough for real-time inference on a laptop-class CPU. The landmark sequence is passed to a CNN-LSTM classifier that recognizes both static hand shapes and dynamic gesture trajectories, addressing the continuous-signing limitation identified in Section II. Recognized signs are buffered and assembled into sentences, then rendered as on-screen text and converted to speech.

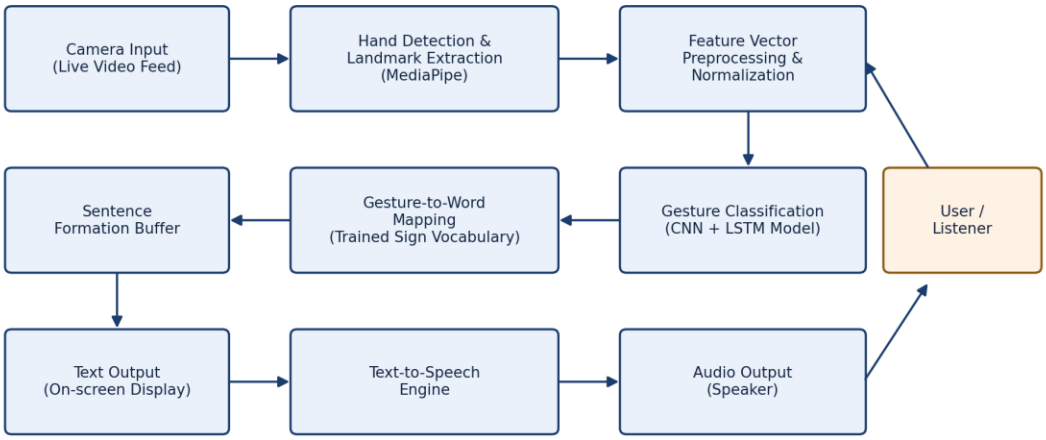


Fig. 1. Block diagram of the proposed real-time sign language detection and translation pipeline.

B. Hardware Requirements

- Standard RGB webcam (720p or higher, 30 fps) — no depth sensor or specialized glove required.
- Laptop or desktop computer with a minimum quad-core CPU; a mid-range GPU (e.g., NVIDIA GTX/RTX class) is recommended for training but not required for inference.
- Speaker or audio output device for the text-to-speech module.

C. Software Requirements

- Python 3.x as the primary development language.
- OpenCV for video capture and image preprocessing.
- MediaPipe (or equivalent) for real-time hand landmark detection.
- TensorFlow / Keras or PyTorch for building and training the CNN-LSTM classification model.
- pyttsx3 or a cloud text-to-speech API for speech synthesis.
- Streamlit or a lightweight Flask/PyQt interface for the user-facing application.

D. Working Principle / Algorithm

- Step 1 — Capture live video frames from the webcam and detect the presence of one or two hands in frame.
- Step 2 — Extract normalized hand landmark coordinates per frame and buffer a short temporal window of frames to capture motion.

- Step 3 — Feed the landmark sequence to the trained CNN-LSTM model, which outputs a probability distribution over the sign vocabulary; apply a confidence threshold to reject uncertain predictions.
- Step 4 — Append high-confidence predictions to a rolling sentence buffer, apply basic grammar smoothing, and simultaneously render the result as on-screen text and speech output.

IV. System Design

The system is organized into three logical layers: (1) an acquisition and perception layer responsible for video capture and hand landmark extraction, (2) a recognition layer that performs gesture classification and temporal sequence modeling, and (3) an output layer that converts recognized signs into text and speech. Fig. 1 in Section III illustrates the data flow between these layers. Additional design artifacts — including the CNN-LSTM network architecture diagram, the sentence-assembly state machine, and the user-interface wireframe — will be developed during the design phase and included in the final report as Fig. 2, Fig. 3, and beyond, each referenced explicitly in the surrounding text with a caption placed below the figure.

V. Implementation Plan and Timeline

The project is planned over a 15-week academic timeline, structured as follows:

Phase / Task	Start Week	End Week	Deliverable
Literature review & requirement analysis	1	2	Finalized requirements & related-work summary
System design & architecture finalization	3	3	Architecture diagrams & dataset plan
Dataset collection / procurement & tool setup	4	5	Labeled gesture dataset & configured dev environment
Core implementation — landmark extraction & model training	6	9	Trained CNN-LSTM classifier
Core implementation — sentence assembly & TTS integration	10	11	End-to-end functional prototype
Integration & testing	12	13	Accuracy, latency & usability test results
Documentation & final report	14	15	Final report & project demonstration

VI. Expected Outcomes

- A working software prototype that detects hand signs from a live webcam feed and outputs corresponding text and synthesized speech in real time.
- Gesture classification accuracy of at least 90% on a held-out validation set, with end-to-end latency under 300 ms per recognized sign.
- A documented, labeled gesture dataset and trained model that can be extended to a larger sign vocabulary in future work.
- Improved accessibility for Deaf and hard-of-hearing individuals in everyday settings such as classrooms, reception desks, and service counters, and a reduced dependency on human interpreters for routine interactions.

VII. Feasibility and Cost Analysis

Technical feasibility is high: all core components — video capture, hand landmark detection, and CNN/LSTM model training — rely on widely available open-source libraries (OpenCV, MediaPipe, TensorFlow/PyTorch) and standard consumer hardware, all within the team's existing skill set. Economic feasibility is favorable, since no specialized sensors or proprietary hardware are required; the primary cost is optional cloud GPU time for faster training, which can be substituted with local hardware at the expense of training time. Time feasibility fits comfortably within a standard academic semester, as reflected in the 15-week plan in Section V.

Item	Purpose	Estimated Cost
Webcam (720p/1080p)	Video input for gesture capture	[₹ / $ amount]
Cloud GPU credits (training only)	Accelerated model training	[₹ / $ amount]
Miscellaneous (storage, TTS API credits)	Data storage & speech synthesis calls	[₹ / $ amount]
Total (approx.)		[₹ / $ amount]

VIII. Conclusion

This proposal presents a real-time, camera-only sign language detection and translation system designed to bridge the communication gap between Deaf or hard-of-hearing individuals and those unfamiliar with sign language. By combining efficient hand landmark extraction with a compact CNN-LSTM classifier and integrated text-to-speech output, the proposed system targets high recognition accuracy and low latency without relying on specialized wearable hardware. Completing this project will yield a functional

prototype, a labeled gesture dataset, and quantitative performance benchmarks, offering a practical and affordable step toward more inclusive, everyday communication technology.

IX. References

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